

Control Signals			Functions for					
cs2	cs1	cs0	Group 1	Group 2	Group 3	Group 4	Group 5	Group 6
0	0	1	Increment A	Sub	Add with carry	Sub with borrow	Transfer A	Transfer A
0	1	1	Transfer A	Sub with borrow	Add	Decrement A	Add	Sub with borrow
1	0	1	Decrement A	Add	Decrement A	Sub	Sub with borrow	Add
1	1	1	Add with carry	Transfer A	Transfer A	Add	Add with carry	Add with carry
X	0	0	AND	NOT	XOR	XOR	XOR	OR
X	1	0	OR	OR	OR	NOT	AND	NOT

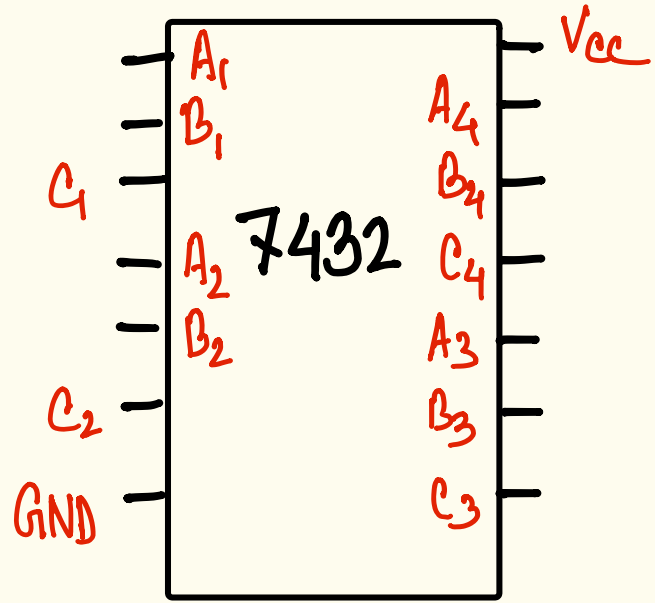
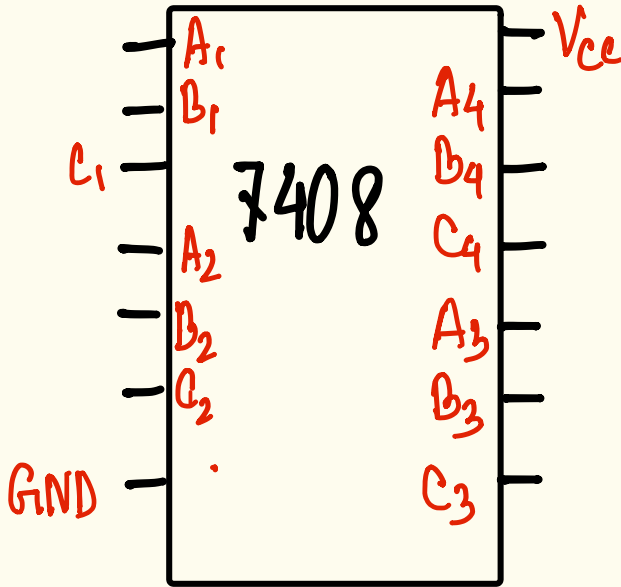
AND → 7408

OR → 7432

NOT → 7404

Adder → 7483

S_2	S_1	S_0	F	X_i	Y_i	C_{in}
0	0	1	A	A	0	0
0	1	1	A+B	A	B	0
1	0	1	A-B-1	A	$\bar{B}$	0
1	1	1	A+B+1	A	B	1
X	0	0	$A \oplus B$	A	B	0
X	1	0	AB	$A + \bar{B}$	$\bar{B}$	0



$Y_i \Rightarrow$

$Z = S_0 C_{in}$

$S_2 \backslash S_1 S_0$	00	01	11	10
00				1
01		1		1
11	1		1	
10	1		1	

$C_{in} = S_2 S_1 S_0$

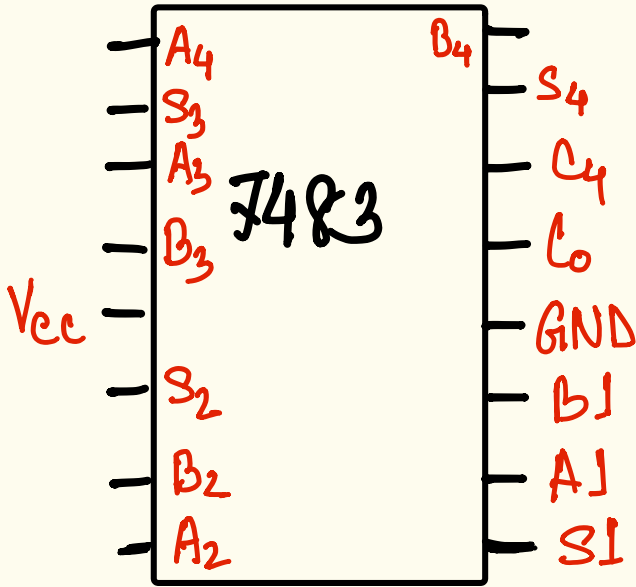
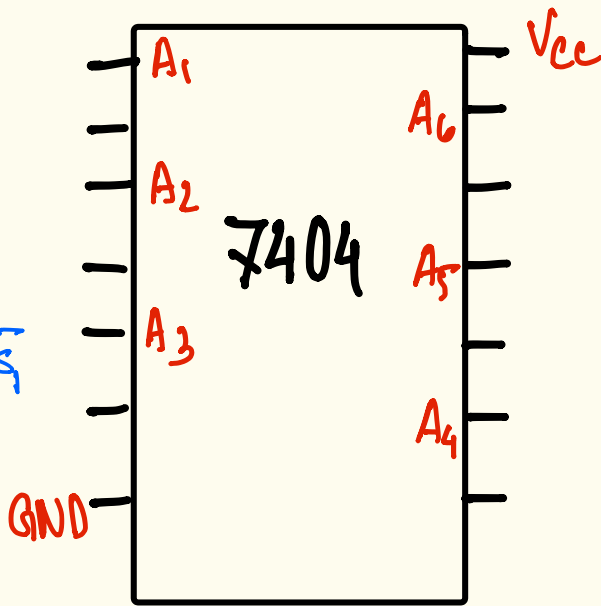
$Y_i = B_i (S_1 \odot S_0) + \bar{B}_i (S_1 \bar{S}_0 + S_2 S_0 \bar{S}_1)$

$X_i \Rightarrow$

$X_i = A_i + \bar{B}_i \cdot S_1 \cdot \bar{S}_0$

S_2	S_1	S_0	Y_i
0	0	1	0
0	1	1	$B \rightarrow \bar{S}_2 S_1 S_0$
1	0	1	$\bar{B} \rightarrow S_2 \bar{S}_1 S_0$
1	1	1	$B \rightarrow S_2 S_1 S_0$
0	0	0	$B \rightarrow \bar{S}_2 \bar{S}_1 \bar{S}_0$
1	0	0	$B \rightarrow S_2 \bar{S}_1 \bar{S}_0$
0	1	0	$\bar{B} \rightarrow \bar{S}_2 S_1 \bar{S}_0$
1	1	0	$\bar{B} \rightarrow S_2 S_1 \bar{S}_0$

$S_0 \bar{S}_1 S_2 + \bar{S}_0 S_1$
 $S_0 S_1 + \bar{S}_0 \bar{S}_1$



Circuit Diagram:

$S_2, S_1, S_0, A_3, A_2, A_1, A_0, B_3, B_2, B_1, B_0$

$$Y_i = B_i (S_1 \odot S_0) + \overline{B_i} (S_1 \overline{S_0} + S_0 \overline{S_1})$$

$$= B_i (\overline{S_1} \overline{S_0} + S_1 S_0) + \overline{B_i} (S_1 \overline{S_0} + S_0 \overline{S_1})$$

$$X_i = A_i + \overline{B_i} S_i \overline{S_j}$$

$C \vee C_{in}$

$$\begin{array}{r} 9 \\ 12 \\ +1 \\ \hline 22 \end{array}$$

$$\begin{array}{r} S \vee C Z \\ \hline 0 \ 1 \ 1 \ 0 \end{array}$$

$$\begin{array}{r} 10 \\ 13 \\ \hline 23 \\ 10111 \end{array}$$

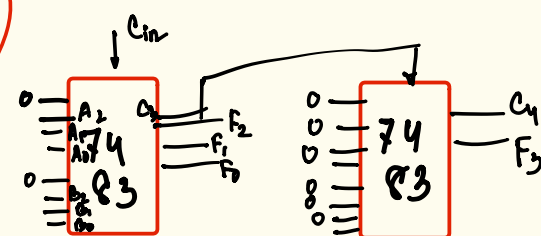
$$\begin{array}{r} 1001 \\ 1101 \\ \hline \end{array}$$

$$\begin{array}{r} 1100 \end{array}$$

$$\begin{array}{r} 10110 \\ 1111 \\ \hline 11001 \end{array}$$

$$\begin{array}{r} 11001 \\ 1101 \\ \hline 10110 \end{array}$$

$$F_2 \times B$$



7408

7408

7408

7408

74 32

74 32

74 32

74 32

74 32

7408

7408

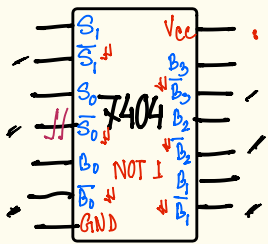
7408

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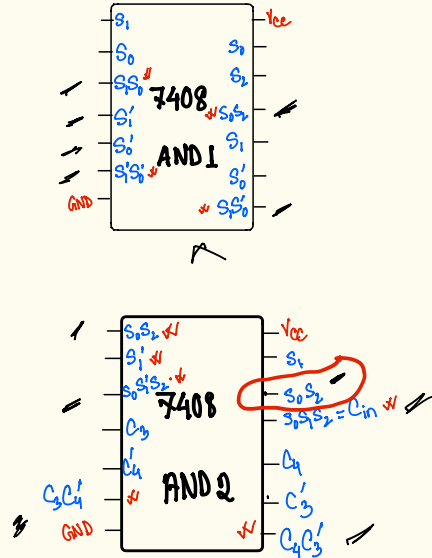
7404

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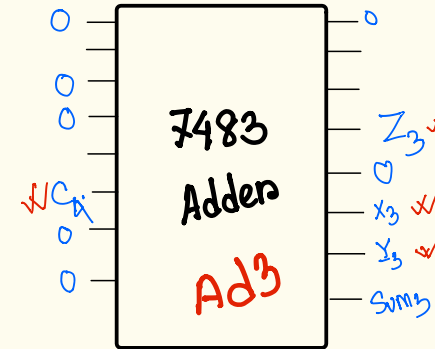
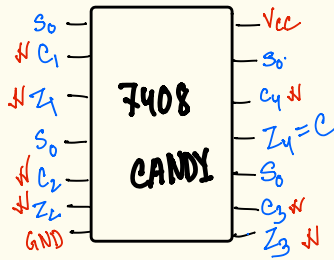
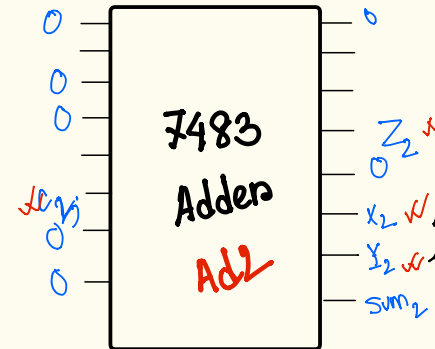
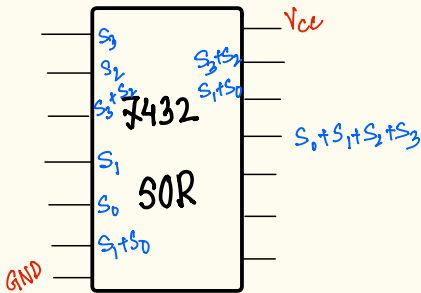
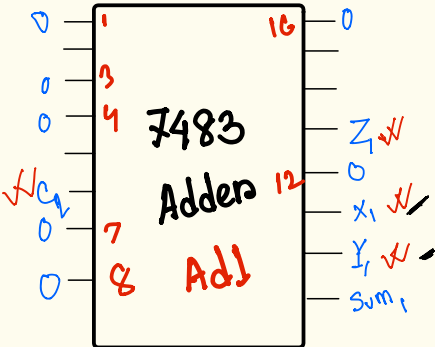
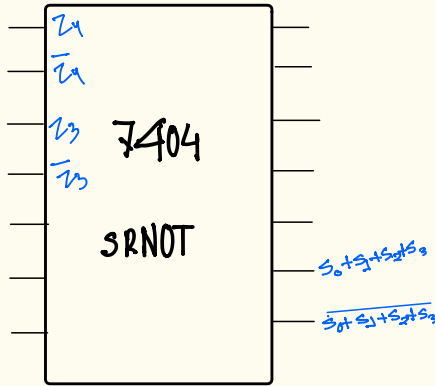
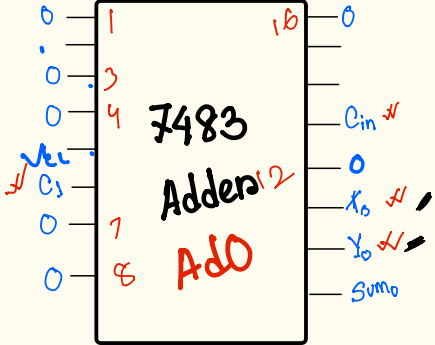
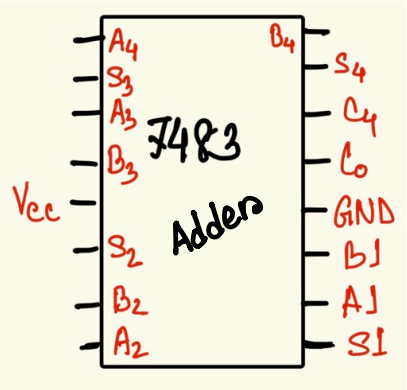
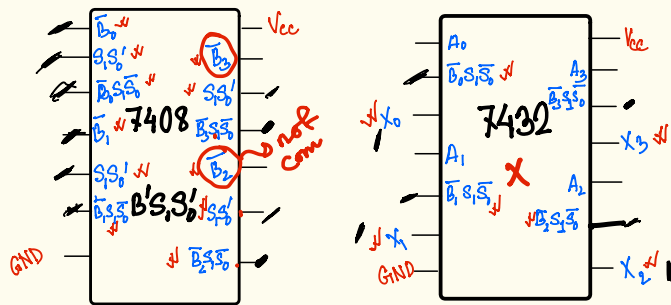
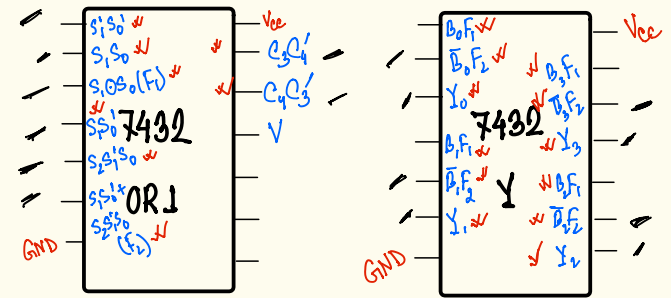
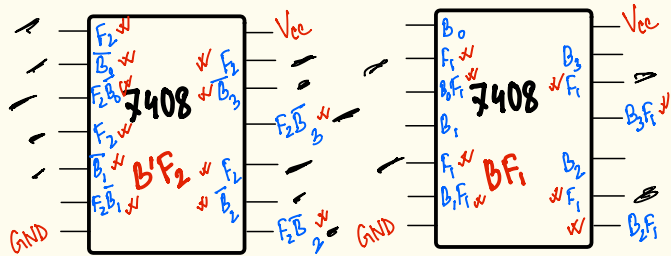
Fault Circuit



X



X



Addr0

Ad1

Ad2

Ad3

s2	s1	s0
----	----	----

b3	b2	b1	b0
----	----	----	----

Candy

γ

$\times$

A_0 ☐

A_1 ☐

A_2 ☐

A_3 ☐

not-1

AND-1

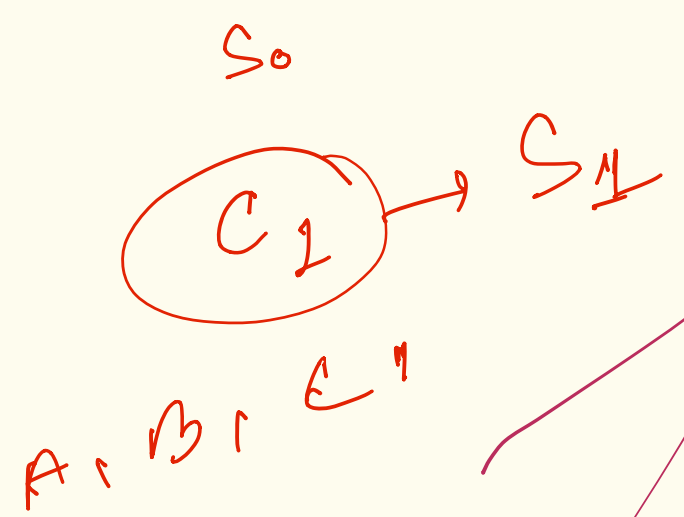
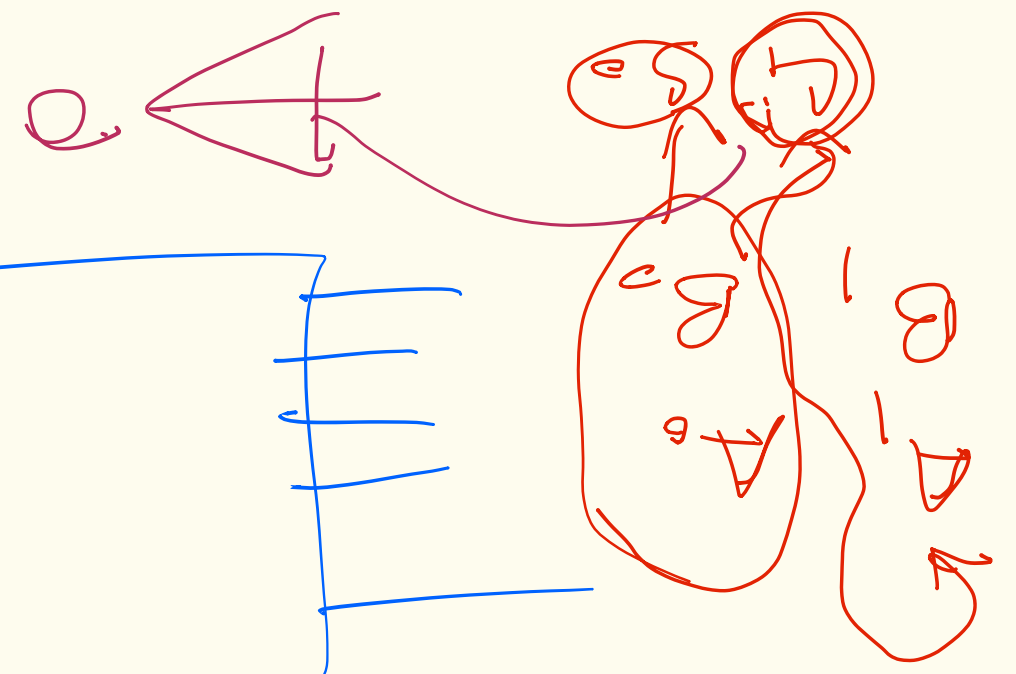
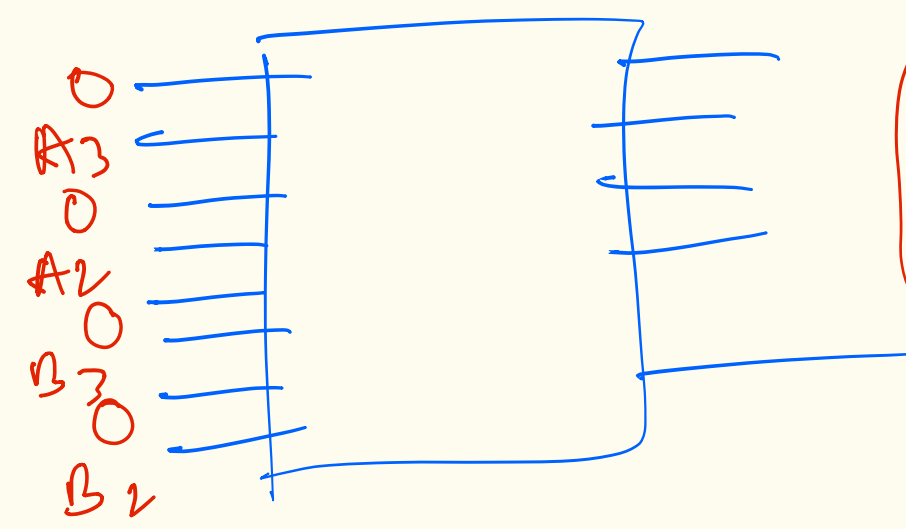
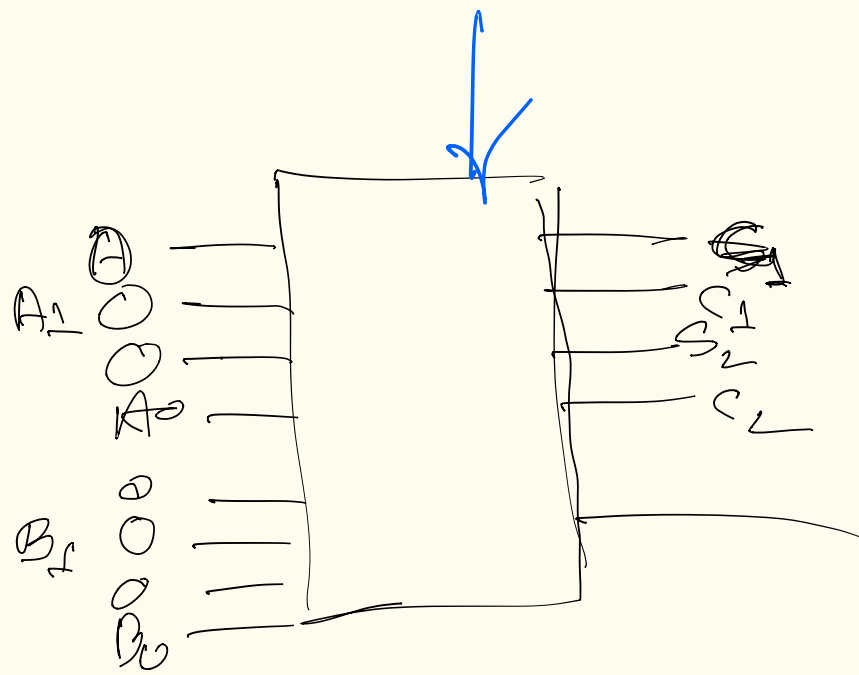
OR-1

AND-2

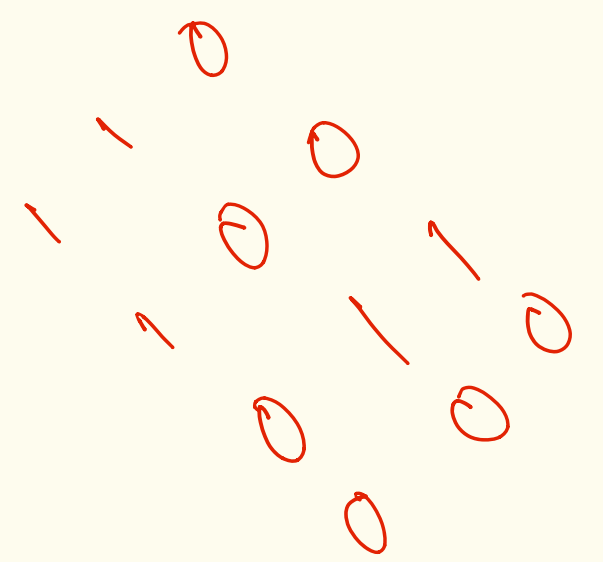
$B'F_2$

$B'S_1S_0$

BF_1



Signature



101 → 0101 Sub with Z borrow

$$\begin{array}{r} 0101 \\ - 0011 \\ \hline 0010 \\ 1111 \\ \hline 1(0)001 \end{array}$$

S	V	C	Z	
0	0	1	0	1000
0	1	1	0	<u>0001</u>
				0111
				<u>1111</u>

$$\begin{array}{r} 0111 \\ 0111 \\ \hline 0000 \\ 1111 \\ \hline 1111 \end{array}$$

$$\begin{array}{r} S V C Z \\ \hline 1000 \end{array}$$

001 → For 001 → Transfer A
 ↳ No C & V

$$\begin{array}{r} 1100 \\ 0101 \\ \hline 1001 \end{array}$$

S	V	C	Z
<u>1</u>	<u>0</u>	<u>0</u>	<u>0</u>

$$\begin{array}{r} 1100 \\ 0101 \\ \hline 1100 \end{array}$$

<u>S</u>	<u>V</u>	<u>C</u>	<u>Z</u>
1	0	0	0

Logical Operation

No Cout & V

010 → AND

1	1	1	1
1	0	0	1
1	0	0	1
<u>s</u>	<u>v</u>	<u>e</u>	<u>z</u>
1	0	0	0

0	1	1	1
0	1	1	1
1	1	1	0

✓ ✓

0 1

s	v	e	z
1	1	0	0

011 → Add

0	1	1	1
0	0	0	1
1	0	0	0
s	v	e	z
1	1	0	0

Handwritten notes: C_4 (with arrow to the first 1), C_3 (with arrow to the first 0), C_2 (with arrow to the second 0), C_1 (with arrow to the last 0).

1000			
1000			
100000			
s	v	e	z
0	1	1	1

←

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0	1	1	Transfer A	Sub with borrow	Add	Decrement A	Add	Sub with borrow
1	0	1	Decrement A	Add	Decrement A	Sub	Sub with borrow	Add
1	1	1	Add with carry	Transfer A	Transfer A	Add	Add with carry	Add with carry
X	0	0	AND	NOT	XOR	XOR	XOR	OR
X	1	0	OR	OR	OR	NOT	AND	NOT

$$\begin{array}{r} \text{S V C Z} \\ \hline 0 \ 0 \ 0 \ 1 \end{array}$$

$$\begin{array}{r} 111 \\ \hline 0101 \\ 1011 \\ \hline 10000 \\ +1 \\ \hline 10001 \end{array}$$

$$\begin{array}{r} \text{S V C Z} \\ \hline 0 \ 0 \ 1 \ 0 \end{array}$$

$$\begin{array}{r} 101 \\ \hline 0101 \\ 1011 \\ \hline 11010 \\ 1111 \\ \hline 1001 \end{array}$$

$$\begin{array}{r} \text{S V C Z} \\ \hline 1 \ 1 \ 0 \ 0 \end{array}$$

$$\begin{array}{r} 010 \\ \hline 0110 \\ 1001 \\ \hline 0000 \end{array}$$

$$\begin{array}{r} 0101 \\ 1011 \\ \hline 0001 \\ \text{S V C Z} \\ \hline 0 \ 0 \ 0 \ 0 \end{array}$$

$$\begin{array}{r} 011 \\ \hline 0101 \\ 1011 \\ \hline 10000 \\ \text{S V C Z} \\ \hline 0 \ 0 \ 1 \ 1 \end{array}$$

Overflow flag indicates that arithmetic operation has exceeded the capacity of ALU's bit width.

C $\rightarrow$ For unsigned arithmetic operation (Crosses 15)

V $\rightarrow$ For signed arithmetic operation (+7 to -8)

Valid Range for V $\rightarrow$ -8 to +7

$$\hookrightarrow \bar{A}_3 \bar{B}_3 S_3 + A_3 B_3 \bar{S}_3$$

$$= (A_3 \oplus B_3) \oplus S_3$$

$$V = C_3 \oplus C_4 \rightarrow \text{Carry out of MSB}$$

$\swarrow$
Carry into MSB

$$5 + 4 = 9 \rightarrow \begin{array}{c} \underline{1001} \\ \rightarrow > 7 \end{array} \quad \text{MSB} = 1 \rightarrow \text{2's complement} \quad 1110 + 1 = 1111$$

4 bit $\rightarrow$ 1 bit for sign $\rightarrow$ max value +7

for min value $\rightarrow$ sign bit must be 1